ANCHAL PANDEY

Software Engineer | Full Stack and Backend | AI Applications

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PROFESSIONAL SUMMARY

Software Engineer with 2 years of experience in full stack development and backend engineering, building AI-powered applications for education. Works across Node.js, Express.js, React.js, MongoDB, and MySQL to deliver secure REST APIs with JWT and RBAC, and applies OpenAI, Gemini, and LangChain to build RAG (Retrieval-Augmented Generation) systems and LLM-driven features. Improved API performance by **80%** in a prior backend role.

PROFESSIONAL HIGHLIGHTS

- **Product delivery:** contributed to 11 products in production across AI platforms, learning management, teacher and learner tools, workflow automation, and device management.
- **Backend engineering:** REST APIs and services on Node.js, Express.js, MongoDB, and MySQL, secured with JWT authentication and role-based access control (RBAC).
- **RAG systems:** built EIR, an end-to-end pipeline (ingestion, embeddings, FAISS vector search, retrieval) that grounds LLM responses in source content.
- **AI-powered applications:** created LPG, A3, and Mashy with OpenAI, Gemini, LangChain, and speech APIs, covering lesson generation, adaptive assessment, and voice-first tutoring.
- **Full stack engineering:** React.js and Tailwind CSS interfaces integrated with backend services and AI pipelines.

TECHNICAL SKILLS

Languages: JavaScript, Python, C++, SQL

Frontend: React.js, Tailwind CSS, HTML5, CSS3, Bootstrap

Backend: Node.js, Express.js, REST APIs, JWT Authentication, RBAC, MVC Architecture, Middleware, Caching, Pagination

Databases: MongoDB (Aggregation), MySQL

Vector Search: FAISS, Embeddings

AI Applications: Generative AI, OpenAI, Gemini, LangChain, RAG (Retrieval-Augmented Generation), LLM Applications, AI Agents, Prompt Engineering, Speech-to-Text

Tools and Cloud: Git, GitHub, Postman, Docker, AWS S3, Moodle LMS Integrations

Engineering: Full Stack Development, Backend Engineering, System Design, Data Structures and Algorithms

PROFESSIONAL EXPERIENCE

Software Engineer (Full Stack and AI) | Mashmari Consultants Pvt Ltd **Feb 2025 – Present**

- Developed backend services and REST APIs with Node.js, Express.js, MongoDB, and MySQL for AI-driven education products, working from design through release.
- Engineered **EIR**, an ingestion and retrieval pipeline (book parsing, chapter detection, embeddings, FAISS vector search), enabling RAG that grounds LLM responses in source content.
- Created **LPG** with OpenAI, Gemini, and LangChain to generate curriculum-aware lesson plans with PDF export, and **A3** to generate adaptive assessments with learner competency profiling, supporting teacher lesson-planning and assessment workflows.
- Implemented **Mashy**, a voice-first AI tutor, integrating speech APIs with an agent-driven tutoring flow to enable spoken learning for Hindi-speaking students.
- Built React.js and Tailwind CSS interfaces integrated with backend and AI services, and contributed to **CMS, TRP, LRP** and **Moodle LMS** integrations for content publishing, teacher and learner tools, and course management.
- Delivered **Tracker** (activity analytics dashboard), **Doot V5** (autonomous remote-classroom player), and **Netrakshak** (device monitoring and policy management), covering workflow automation and device-management use cases.

Software Developer Intern (Backend) | Innobles Smart Technologies **Jul 2024 – Jan 2025**

- Designed secure REST APIs with Node.js, Express.js, and MongoDB using MVC architecture and reusable middleware, giving the team a maintainable codebase with consistent validation, error handling, and logging.
- Implemented JWT authentication and RBAC for Admin, Super Admin, and Agent roles, enforcing role-scoped access to protected resources.
- Improved API response performance by **80%** using caching strategies and paginated queries.
- Produced ticket-trend and agent-performance reports with MongoDB Aggregation pipelines, and added per-user activity logs that sped up debugging and audits.

Earlier Internships | Ricoz Pvt. Ltd. (MERN Developer), CrazyWeb Technology (Frontend) **Aug 2023 – Jun 2024**

EDUCATION

B.Tech, Computer Science and Engineering | Bundelkhand University, Jhansi | CGPA: 8.31 **Nov 2021 – May 2025**

ACHIEVEMENTS

- 500+ DSA problems solved (GeeksforGeeks, CodeChef, HackerRank)
- 5-star C++ on HackerRank
- Selected for IDE Bootcamp (2nd Edition)